

Tensor-Free Calderón Duals for Multi-Norm Littlewood–Paley Families

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Status

This is a candidate full solution, pending human review, of the first question in Section 13, item 4 of Hejna–Nagel–Ricci [1]. It proves the intrinsic tensor-free conclusion of Section 4.2: existence of an associated Calderón reproducing family, with arbitrarily high (indeed infinite) coordinate cancellation. It does not claim the broader L^1 -equivalence question in the following sentence of the source.

Source question

The source is Agnieszka Hejna, Alexander Nagel, and Fulvio Ricci, *Littlewood–Paley square functions and the local Hardy space for Multi-Norm Structures on $\mathbb{R}^d$* , arXiv:2507.09399v1. On page 63, Section 13, item 4, the authors ask whether the results of Section 4.2 continue to hold without assuming that the functions Ψ_L are tensor products, while retaining uniform boundedness in Schwartz norms, cancellation in each marked variable, and

$$\sum_{L \in \mathcal{L}} \Psi_L^{(-L)} = \delta_0.$$

possible using the techniques of Section 5.

3. We do not consider characterizations of $\mathbf{h}_{\mathbf{E}}^1(\mathbb{R}^d)$ via maximal operators. Natural candidates as “multi-norm maximal operators” are those of the form $M_\varphi f(\mathbf{x}) = \sup_{\mathbf{t} \in \Gamma(\mathbf{E})} |f * \varphi^{(-\mathbf{t})}(\mathbf{x})|$, with $\int \varphi = 1$, as well as grand-maximal analogues.
4. We do not know whether the results of Section 4.2 still hold removing the assumption that the Ψ_L are tensor products, but keeping the other conditions, that is, uniform boundedness in Schwartz norms, cancellation in each variable $\mathbf{x}_i$ with $i \in D_S$ and $\sum_{L \in \mathcal{L}} \Psi_L^{(-L)} = \delta_0$. More generally, we do not know whether it is possible to obtain, in a simple form, a larger class of mutually L^1 -equivalent square functions which includes those of both tensor- and convolution-type.
5. In this paper we do not consider continuous-parameter square functions that are natural counter-

Definitions and notation

We use the notation of [1, Sections 2–4]. Thus a standard matrix $\mathbf{E}$ determines an index set $\mathcal{L} \subset \mathbb{N}^n$, a finite partition $\mathcal{L} = \bigcup_S \mathcal{L}_S$ by marked partitions, and a set D_L of marked coordinates for each $L \in \mathcal{L}$. If $L = (\ell_1, \dots, \ell_n)$, write

$$2^{-L}\xi = (2^{-\ell_1}\xi_1, \dots, 2^{-\ell_n}\xi_n),$$

with the fixed homogeneous dilations inside each coordinate block understood. The Fourier transform of $\Psi_L^{(-L)}$ is therefore $\widehat{\Psi}_L(2^{-L}\xi)$.

For $L \in \mathcal{L}_S$, where $S = \{(A_r, k_r) : 1 \leq r \leq s\}$, the geometry established in [1, Section 3.4] gives

$$\ell_j = \left\lfloor \frac{\ell_{k_r}}{e(k_r, j)} \right\rfloor \quad (j \in A_r), \quad D_L = \{k_1, \dots, k_s\}. \quad (1)$$

In particular, on each of the finitely many pieces $\mathcal{L}_S$, all unmarked coordinates of L are determined by the marked ones.

Cancellation in the variable x_i means

$$\int_{\mathbb{R}^{d_i}} \Psi_L(x_1, \dots, x_i, \dots, x_n) dx_i = 0$$

as a Schwartz function of the remaining variables. Equivalently, $\widehat{\Psi}_L$ vanishes on the hyperplane $\{\xi_i = 0\}$.

Proof intuition

The partition identity prevents all frequency-local atoms from being small at the same point. The only possible complication is that a general Schwartz atom is not frequency supported in its nominal multi-norm tube. Cancellation controls leakage below a marked scale, Schwartz decay controls leakage above any scale, and the marked-partition geometry turns the total leakage into ordinary geometric series. After enlarging the tubes by a fixed amount, the atoms associated with tubes missing a given frequency have total magnitude less than $1/2$.

We then keep only the atoms in those enlarged tubes and form their pointwise Gram symbol

$$A(\xi) = \sum_L \theta_L(\xi) \left| \widehat{\Psi}_L(2^{-L}\xi) \right|^2.$$

It is bounded below because the retained atoms still sum to at least $1/2$, and only boundedly many are retained. The canonical frame inverse $\theta_L \widehat{\Psi}_L / A$ gives an exact reproducing identity. The cutoffs have fixed compact support after rescaling by L and avoid the marked coordinate hyperplanes. Consequently the duals are uniformly Schwartz and have vanishing moments of every order in every marked variable.

The adapted tail lemma

Choose smooth homogeneous gauges ρ_i on the frequency blocks $\mathbb{R}^{d_i}$. Their precise choice is immaterial.

Lemma 1 (localized tails and overlap). *Let $\{\Psi_L : L \in \mathcal{L}\}$ be uniformly bounded in every Schwartz seminorm and suppose that $\widehat{\Psi}_L$ vanishes on $\{\xi_i = 0\}$ whenever $i \in D_L$. Given $\varepsilon > 0$, there is $R > 1$ and, for every $D \subseteq \{1, \dots, n\}$, a function $\vartheta_D \in C_c^\infty(\mathbb{R}^d)$ with $0 \leq \vartheta_D \leq 1$ such that the following hold. Put*

$$a_L(\xi) = \widehat{\Psi}_L(2^{-L}\xi), \quad \vartheta_L(\xi) = \vartheta_{D_L}(2^{-L}\xi).$$

Then

1. $\vartheta_D = 1$ when

$$2^{-R} \leq \rho_i(\eta_i) \leq 2^R \quad (i \in D), \quad \rho_i(\eta_i) \leq 2^R \quad (i \notin D),$$

and its support is contained in the same set with R replaced by $R + 1$. In particular, its support misses $\{\eta_i = 0\}$ for every $i \in D$.

2. Uniformly in ξ ,

$$\sum_{L: \vartheta_L(\xi) \neq 1} |a_L(\xi)| < \varepsilon. \quad (2)$$

3. At every ξ , at most N_R of the functions $\vartheta_L(\xi)$ are nonzero, where N_R is independent of ξ .

4. If $\vartheta_L(\xi)\vartheta_P(\xi) \neq 0$ for some ξ , then $|L - P| \leq C_R$.

Proof. Smooth cutoffs with property (1) are obtained from fixed annular cutoffs in the marked variables and fixed ball cutoffs in the unmarked variables.

We give the estimates behind (2)–(4). Uniform Schwartz bounds and repeated Hadamard factorization in the marked coordinate blocks imply, for every N ,

$$|\widehat{\Psi}_L(\eta)| \leq C_N \prod_{i \in D_L} \min\{1, \rho_i(\eta_i)\} \left(1 + \sum_{j=1}^n \rho_j(\eta_j)\right)^{-N}. \quad (3)$$

Indeed, vanishing on each marked hyperplane supplies one factor from that coordinate block, and the remaining factors are uniformly Schwartz.

Fix a marked partition $S = \{(A_r, k_r) : 1 \leq r \leq s\}$ and sum (3), with $\eta = 2^{-L}\xi$, over $L \in \mathcal{L}_S$. By (1), this is a sum only over the marked integers $\ell_{k_1}, \dots, \ell_{k_s}$. If $2^{-\ell_{k_r}} \rho_{k_r}(\xi_{k_r}) < 2^{-R}$, the cancellation factor gives geometric decay. If a marked or unmarked normalized frequency exceeds 2^R , the last factor in (3) gives geometric decay; take N larger than the number of free marked indices. Summing successively in the s marked integers gives

$$\sup_{\xi} \sum_{L \in \mathcal{L}_S: \vartheta_L(\xi) \neq 1} |a_L(\xi)| \leq C_S 2^{-c_S R}.$$

When $\xi_{k_r} = 0$, the corresponding summands vanish and the same estimate is immediate. There are only finitely many marked partitions, so increasing R proves (2).

If $\vartheta_L(\xi) \neq 0$, every marked ℓ_{k_r} lies in an interval of length $O(R)$ determined by $\rho_{k_r}(\xi_{k_r})$. Formula (1) then determines all unmarked components. This gives the uniform cardinality bound in (3).

Finally, the supports of the ϑ_L are fixed enlargements of the frequency tubes B_L of [1, Section 3.4]. Thus a common point of two supports gives an intersection of fixed enlargements of the two tubes. The fixed-enlargement version of [1, Lemma 3.8(iv)] gives $|L - P| \leq C_R$. Equivalently, this also follows directly by applying (1) to the finitely many possible pairs of marked partitions. This proves (4). $\square$

Main theorem

Theorem 1 (tensor-free Calderón dual). *Let $\mathbf{E}$ be a standard matrix and let $\mathcal{L}, D_L$ be the associated multi-norm index data from [1]. Suppose $\{\Psi_L : L \in \mathcal{L}\} \subset \mathcal{S}(\mathbb{R}^d)$ satisfies:*

1. *the family is uniformly bounded in every Schwartz seminorm;*

2. Ψ_L has cancellation in x_i for every $i \in D_L$;

3. in the sense of tempered distributions,

$$\sum_{L \in \mathcal{L}} \Psi_L^{(-L)} = \delta_0.$$

Then there is a family $\{\tilde{\Psi}_L : L \in \mathcal{L}\} \subset \mathcal{S}(\mathbb{R}^d)$, uniformly bounded in every Schwartz seminorm, such that

$$\sum_{L \in \mathcal{L}} \Psi_L^{(-L)} * \tilde{\Psi}_L^{(-L)} = \delta_0 \quad (4)$$

in $\mathcal{S}'(\mathbb{R}^d)$. Moreover, for every $i \in D_L$, $\tilde{\Psi}_L$ has vanishing moments of every order in the variable x_i . In particular it has cancellation of any prescribed finite order.

Proof. Apply Lemma 1 with $\varepsilon = 1/2$. The estimate used in its proof also gives absolute locally uniform convergence of $\sum_L a_L(\xi)$, including all derivatives after lowering the decay order. The assumed distributional identity therefore implies the pointwise identity

$$\sum_{L \in \mathcal{L}} a_L(\xi) = 1. \quad (5)$$

Let

$$\mathcal{C}(\xi) = \{L : \vartheta_L(\xi) = 1\}.$$

By (2) and (5),

$$\left| \sum_{L \in \mathcal{C}(\xi)} a_L(\xi) \right| \geq \frac{1}{2}.$$

Lemma 1(3) now gives

$$\max_{L \in \mathcal{C}(\xi)} |a_L(\xi)| \geq \frac{1}{2N_R}.$$

Define the localized frame symbol

$$A(\xi) = \sum_{L \in \mathcal{L}} \vartheta_L(\xi) |a_L(\xi)|^2. \quad (6)$$

The sum is locally finite by Lemma 1(3), and

$$A(\xi) \geq (2N_R)^{-2} > 0 \quad (7)$$

for every ξ .

For $L \in \mathcal{L}$, define

$$\widehat{\Psi}_L(\eta) = \frac{\vartheta_{D_L}(\eta) \overline{\widehat{\Psi}_L(\eta)}}{A(2^L \eta)}. \quad (8)$$

We first verify uniform Schwartz bounds. The numerator in (8) has support in one of finitely many fixed compact sets. On such a support, put $\xi = 2^L \eta$. A term indexed by P can occur in $A(\xi)$ only when $\vartheta_P(\xi) \vartheta_L(\xi) \neq 0$. Lemma 1(4) then gives $|P - L| \leq C_R$, and Lemma 1(3) bounds the number of such terms. After returning to the normalized variable η , all relative dilations 2^{L-P} range over a fixed compact set. Uniform Schwartz bounds on the Ψ_P consequently give uniform bounds for

every derivative of $A(2^L\eta)$ on the support of the numerator. Together with (7) and the quotient rule, this proves that the functions in (8) are uniformly bounded in C_c^∞ . Their inverse Fourier transforms are therefore uniformly bounded in every Schwartz seminorm.

For $i \in D_L$, the support of ϑ_{D_L} misses the entire hyperplane $\{\eta_i = 0\}$. Hence $\widehat{\Psi}_L$ and all its derivatives vanish there. This is precisely cancellation of every order in x_i .

Finally,

$$\widehat{\widetilde{\Psi}_L^{(-L)}}(\xi) = \frac{\vartheta_L(\xi)\overline{a_L(\xi)}}{A(\xi)}.$$

Therefore, by (6),

$$\sum_{L \in \mathcal{L}} \widehat{\Psi}_L^{(-L)}(\xi) \widehat{\widetilde{\Psi}_L^{(-L)}}(\xi) = \frac{\sum_L \vartheta_L(\xi) |a_L(\xi)|^2}{A(\xi)} = 1.$$

The sums are locally finite after insertion of the cutoffs, so this identity holds in $\mathcal{S}'(\mathbb{R}^d)$. Taking inverse Fourier transforms proves (4). $\square$

Verification notes and limitations

- The conjugate in (8) is essential for complex families: it makes the denominator a positive Gram symbol.
- The case $L = 0$, where $D_L = \emptyset$, uses only a fixed low-frequency ball cutoff. All nonzero boundary and interior indices have at least one marked coordinate, so local finiteness is unchanged.
- The only geometric input beyond the definitions is the scale relation (1) and bounded overlap of fixed tube enlargements, both consequences of Section 3 of the source paper.
- The main reviewer focus should be Lemma 1, especially the uniform summation over different marked partitions. The rest of the proof is an exact algebraic frame inversion.
- The theorem supplies the associated Calderón family, which is the tensor-free content of Section 4.2. It does not assert the tensor-specific partial-sum identity involving the separately defined functions Φ_L , nor does it by itself prove the broader L^1 square-function equivalence asked in the next sentence of the source.

Novelty check

On July 21, 2026, bounded searches were performed for the exact question sentence, the source identifier and title, “multi-norm Calderón reproducing formula non-tensor,” and “multi-norm Littlewood-Paley family.” Searches also used the relevant author/topic combinations. They found the source preprint and adjacent Littlewood–Paley literature, but no later paper or source revision answering this question. This is not an exhaustive bibliographic proof of novelty; novelty confidence is moderate.

References

- [1] A. Hejna, A. Nagel, and F. Ricci, *Littlewood-Paley square functions and the local Hardy space for Multi-Norm Structures on $\mathbb{R}^d$* , arXiv:2507.09399v1 (2025).